

1. Administrative & Study Identification

**Full Study Title:** A Phase III, Multicenter, Randomized, Double-Blind, Placebo-Controlled, Parallel-Group Efficacy and Safety Study of Trontinemab in Participants With Early Symptomatic Alzheimer's Disease (MCI to Mild Dementia Due to AD) **Short Title/Acronym:** TRONTIER 2 **Protocol Number:** WN45447 **NCT Number:** NCT07170150 **Sponsor Name:** Hoffmann-La Roche **Investigational Product:** Trontinemab (R07126209) **Phase of Development:** Phase III **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

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2. Study Synopsis & Rationale

**2.1 Study Objective** **Primary Objective:** To assess the efficacy and safety of trontinemab in participants with early symptomatic Alzheimer's disease (AD), specifically mild cognitive impairment (MCI) to mild dementia due to AD. **Secondary Objectives:** To evaluate the reduction in amyloid plaques, and to assess changes in secondary cognitive/functional measures including the Mini-Mental State Examination (MMSE) and the Alzheimer's Disease Assessment Scale-Cognitive Subscale (ADAS-Cog13). To monitor safety parameters, particularly the incidence and severity of amyloid-related imaging abnormalities (ARIA).

**2.2 Background & Rationale** Alzheimer's disease (AD) is a progressive neurodegenerative disorder caused by changes in beta-amyloid and tau proteins in the brain. A key hurdle in developing anti-Aβ monoclonal antibodies has been limited penetration of the blood-brain barrier (BBB). Trontinemab features Brainshuttle technology designed to bind to human transferrin receptor 1 (TfR1) to cross the BBB more effectively, aiming to clear amyloid plaques rapidly while minimizing ARIA events (brain swelling or bleeding).

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3. Study Design & Methodology

**3.1 Design Framework** **Study Type:** Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Parallel-Group

**3.2 Planned Enrollment & Duration** **Target N\$:** 800 participants **Number of Sites:** Approximately 113 to 135 global locations **Estimated Study Start:** 12-Nov-2025 **Estimated Primary Completion:** 07-Jun-2028

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 50 to 90 years with probable AD dementia or MCI due to AD (Alzheimer's clinical syndrome Stage 3 or 4). 2. Screening MMSE score  $\geq 22$  and a Global Clinical Dementia Rating (CDR-GS) of 0.5 or 1.0. 3. Evidence of AD pathological process, as confirmed on an amyloid PET scan, or alternatively via CSF tau181/A $\beta$ 42 ratio. 4. Willingness and ability to complete all study aspects, including MRI, PET/CSF, and neuropsychological testing (with a study partner).

4.2 Exclusion Criteria 1. Any evidence of a condition other than AD that may affect cognition (e.g., vascular dementia, Lewy body dementia, PD). 2. History or presence of clinically significant cerebrovascular disease, severe CNS trauma, or intracranial mass. 3. MRI evidence of significant cerebral abnormalities, including  $\geq 2$  lacunar infarcts,  $\geq 4$  microbleeds, or prior ARIA-E. 4. Inability to tolerate MRI procedures or contraindications to MRI (e.g., incompatible pacemakers).

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Trontinemab 3.6 mg/kg or matching placebo administered monthly for the first 6 months, followed by maintenance doses every 3 months. Route of Administration: Intravenous (IV) infusion. Duration of Treatment: Approximately 1.5 years (up to primary completion).

5.2 Concomitant & Prohibited Therapy Permitted: Standard-of-care medications for stable co-morbidities as approved by the investigator. Prohibited: Other anti-amyloid investigational therapies or monoclonal antibodies targeting AD pathophysiology during the study period.

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6. Study Timeline & Visit Schedule

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|---------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Visit ID            | Window (Days)                                                      | Key Activities/Procedures                                                     | :---                                                                                                       |
| :---                | :---                                                               | <b>Screening</b>                                                              | Day -28 to 0   Informed Consent, MMSE/CDR testing, Amyloid PET/CSF, Baseline MRI                           |
| <b>Baseline</b>     | Day 0                                                              | Randomization, First IV Infusion, Safety Labs                                 | <b>Interim</b>                                                                                             |
| Months 1 to 6       | Monthly IV Infusions, Safety MRI (ARIA monitoring), AE Assessments | <b>Maintenance</b>                                                            | Months 9, 12, 15   Quarterly IV Infusions, Cognitive/Functional Assessments (CDR-SB, ADAS-Cog), PK/PD labs |
| <b>End of Study</b> | Month 18                                                           | Final Cognitive Assessments, Final MRI/PET, Exit Interview, IP Accountability |                                                                                                            |

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoint Definition:** Change from baseline in Clinical Dementia Rating-Sum of Boxes (CDR-SB) score at the end of the treatment period (approx. 18 months). **Measurement Method:** Standardized clinical and caregiver interview conducted by a blinded, certified rater.

**7.2 Secondary & Exploratory Endpoints** \* Change from baseline in MMSE and ADAS-Cog13. \* Proportion of participants achieving amyloid negativity (measured via amyloid PET scan). \* Incidence, severity, and timing of Amyloid-Related Imaging Abnormalities (ARIA-E and ARIA-H) monitored via central MRI review.

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Continuous monitoring at every study visit, with specific emphasis on neurological changes and routine protocol-defined safety MRIs to detect asymptomatic ARIA. **Serious Adverse Events (SAEs):** Standard 24-hour reporting timeline to the sponsor (Hoffmann-La Roche) and local IRBs. **Data Monitoring Committee (DMC):** An independent external Data Monitoring Committee will evaluate unblinded safety data on an ongoing basis.

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Intent-to-Treat (ITT) population for primary efficacy analysis; Safety Set for AE/ARIA evaluation. **Sample Size Justification:** Target enrollment of  $N=800$  is powered to detect a statistically significant treatment difference in the CDR-SB functional score between the trontinemab and placebo arms. **Handling of Missing Data:** Mixed-Model Repeated-Measures (MMRM) and multiple imputation strategies for longitudinal cognitive endpoints.

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Onsite and remote monitoring visits focusing on consent verification, IP accountability, and safety reporting. **Audit Trail:** Centralized reading for all MRIs (for ARIA) and PET scans (for amyloid burden) to ensure blinded, objective assessments.

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## 11. Study Identifiers & Governance

**Primary ID:** WN45447 **Registry Number:** NCT07170150 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** TRONTIER 2 **Official Regulatory Title:** A Phase III, Multicenter, Randomized, Double-Blind, Placebo-Controlled, Parallel-Group Efficacy and Safety Study of Trontinemab in Participants With Early Symptomatic Alzheimer's Disease (MCI to Mild Dementia Due to AD)

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## 12. Clinical Framework (Alzheimer's Disease Specific)

**Primary Condition:** Early Symptomatic Alzheimer's Disease (MCI to Mild Dementia) **Diagnosis Threshold:** Evidence of AD pathology via amyloid PET or CSF (tau181/A $\beta$ 42 ratio); MMSE  $\geq 22$  **Medical Methodology:** Brainshuttle-enabled Anti-Amyloid Monoclonal Antibody Therapy **Optimization Target:** Rapid clearance of brain amyloid plaques with minimized rates of ARIA

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## 13. Study Procedures & Methodology

**Management Pathway:** Stepwise titration and maintenance of IV trontinemab alongside stringent MRI monitoring for ARIA. **Education Protocol (HCP):** Mandatory certification on standardized cognitive assessments (CDR, ADAS-Cog) and ARIA detection/management protocols. **Education Protocol (Patient):** Requirement of a dedicated study partner/caregiver to assist with treatment adherence and cognitive change reporting. **Quality Audit Mechanism:** Centralized biomarker imaging review and rater concordance tracking.

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## 14. Observation & Follow-Up Schedule

**Total Duration:** Approximately 18 months of treatment **Onsite Visit Cadence:** Monthly for the first 6 months, transitioning to every 3 months (quarterly) **Remote Monitoring Frequency:** Inter-visit phone check-ins for AE tracking. **Checklist Review Interval:** Detailed cognitive, functional, and safety evaluation at baseline, mid-point (e.g., 6 months), and end-of-study (approx. 18 months).

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## 15. Sign-off & Approval

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|--------------------------|--------|-----------|---------------|-----|-----|------|-----|
| Role                     | Name   | Signature | Date          | :-- | :-- | :--  | :-- |
| Principal Investigator   | [Name] | _____     | [DD-MMM-YYYY] |     |     |      |     |
| Clinical Operations Lead | [Name] | _____     | [DD-MMM-YYYY] |     |     | Lead |     |
| Statistician             | [Name] | _____     | [DD-MMM-YYYY] |     |     |      |     |